

# myAQM: Interface Portable Air Quality Sensors with Apple Watch

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Figure 1: myAQM's Apple Watch interfaces. (a) & (b): current and historical data on main app, respectively. (c) & (d): two stages of an air quality change notification: short-look (c) followed by long-look (d). (e): PM2.5 & CO<sub>2</sub> complications on the Watch face.

## ABSTRACT

The last decade saw the proliferation of smartwatches as health devices. Coupled with increased ambient air pollution, there is a growing interest in interfacing air quality data with smartwatches. In real-world scenarios, how and through which interaction users benefit from such interfaces is insufficiently investigated. We implement and test myAQM, a ubiquitous computing system which integrates data from a portable air quality monitor into the Apple Watch's interfaces. Early qualitative findings suggest the Watch is capable of providing timely indication of air pollution events and glanceable data. This suggests smartwatches as a potential add-on in communicating air quality data compared to smartphones alone, to be investigated in future studies through quantitative methods.

## CCS CONCEPTS

• Human-centered computing → Ubiquitous and mobile computing systems and tools.

## KEYWORDS

Air quality monitor; Apple Watch; Notifications; Complications

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## 1 INTRODUCTION

Air quality monitoring is important in many real-life situations as anthropogenic pollution continues to rise. Personal air quality monitoring devices have become more popular, providing high-resolution spatio-temporal data at low costs [3]. Notably, the hyper-localization of air quality data like particulate matter smaller than 2.5 $\mu$ m (PM2.5) and carbon dioxide (CO<sub>2</sub>) is insightful as air quality fluctuates depending on proximity to outdoor and indoor pollution sources, allowing one to identify where to go biking or which room needs better ventilation to minimize airborne disease transmissions, for instance.

As wearables have become popular health devices, there is an emerging need to assess how well commercial smartwatches communicate air quality measurements from portable air quality devices through usability testings. Zhong et. al [5] found their Apple Watch prototypes effective at encouraging behaviors to lower CO<sub>2</sub> levels in a controlled office setting. Hosseini et. al [1] tested a Samsung Gear Live app on asthmatic patients, citing their ability to wear the system daily without any burden and the watch's strength when smartphones cannot be used. To our best knowledge, those were among the only studies till now which conducted usability testings on the use of commercial smartwatches for interfacing hyperlocal air quality. To investigate this type of interface in more diverse real-life situations, we implement and user-test myAQM, a ubiquitous computing system that provides Apple Watch users with PM2.5 and CO<sub>2</sub> measurements from a portable air quality monitor.

## 2 MYAQM SYSTEM

### 2.1 Portable air quality monitor

myAQM's air quality monitor uses Sensirion air quality sensors and an ESP32 micro-controller, see Fig. 2(a). This device sends measurements to the myAQM app via Bluetooth Low-Energy (BLE), similar to other low-cost portable air quality monitors [3].



**Figure 2: An overview of the myAQM system. (a): Components of the portable air quality monitor. (b): Data flow between the 3 components: air quality monitor (left), iOS app (center), and watchOS app (right).**

## 2.2 iOS-watchOS myAQM apps

The iOS app serves as the central hub, receiving measurements from the portable device to store in a database and relay data to the Watch, see Fig. 2(b). We use the US Environmental Protection Agency’s Air Quality Index to categorize raw PM2.5 measurements with color coding schemes. For CO<sub>2</sub> measurements, we adopt the following categorization:  $\leq 1,000\text{PPM}$ : good,  $1000\text{--}1500\text{PPM}$ : moderate, and  $>1500\text{PPM}$ : poor [2]. The watchOS app shows current air quality measurements with scales, color coding schemes, and historical data, see Fig. 1(a) & 1(b). As users might find it difficult to associate air quality thresholds with health outcomes and feel helpless when the air quality is not good [3][5], we incorporate possible health effects and suggestions on how to reduce exposure for each category, such as “Decreased cognitive performance & fatigue might be experienced → Consider: ventilate, open windows” (CO<sub>2</sub> Moderate).

**Notification.** Notification is the major way users interact with smartwatches due to its promptness and convenience [4]. When PM2.5 or CO<sub>2</sub> level changes from one category to another, a notification will alert users, see Fig. 1(c), displaying health implications and suggestions on what to do, see Fig. 1(d).

**Complication.** A complication is a widget on the Apple Watch’s face, providing timely data for quick information dissemination through brief “peeks” [4]. myAQM’s complications are simple. Each complication shows the current PM2.5 or CO<sub>2</sub> measurement with a gauge showing the color of the air quality category, see Fig. 1(e).

## 3 USABILITY STUDY

We recruited 9 participants (3 female, 6 male; of which: 4 commuters, 5 on-campus residents) for the study. Each person went through:

- (1) A 9-day usage period of myAQM throughout their daily routines to test the system during real-world use cases
- (2) 90-min semi-structured interviews (before and after usage) to explore their air quality perception, how they used myAQM, and how it might have influenced their awareness and behaviors (especially the Watch)
- (3) Two 20-min surveys during usage period to keep track of participants’ experience to discuss in the 2<sup>nd</sup> interview

## 4 EARLY FINDINGS

Based on the grounded theory, we employed open-coding on the interview transcripts to discover common themes among users. Our initial analysis uncovers two such themes:

First, the Watch’s notifications help users promptly notice air quality changes during daily routines, such as driving (P7), riding the bus (P9), eating out (P4), cooking (P1, P4, P6), going from indoors to outdoors (all participants), and burning candles (P9). Some participants admitted they would not have checked air quality by opening the iOS/watchOS app at all, had it not been for the notifications during the activities. Most participants recalled they would rely on the Watch for at-a-glance data and notifications; they would open the iOS app later for granular data thanks to its bigger screen. This suggests the Watch as a potential add-on in delivering air quality data compared to smartphones alone; however, its quantitative impact remains to be investigated in future comparative studies.

Second, the Watch’s complications act as gateways to the app, as echoed by P5: “When I’m on my watch, since [air quality] is clearly shown on the complications, I just tend to open it and read more about stuff like that” and P7: “Typically I would be glancing [at the air quality complications] when checking the time, weather or a notification.” This suggests the omnipresence of air quality complications: they provide glanceable data and remind the user of the air quality regardless if one intentionally thinks about air quality when looking at the Watch to see the time or other apps.

In future analysis, we will conduct further qualitative analysis with open coding, axial coding, and selective coding to explore if and how these early findings translate to any changes in participants’ awareness and behaviors.

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## REFERENCES

- [1] Anahita Hosseini, Chris M. Buonocore, Sepideh Hashemzadeh, Hannaneh Hojaiji, Haik Kalantarian, Costas Sideris, Alex A.T. Bui, Christine E. King, and Majid Sarrafzadeh. 2017. Feasibility of a Secure Wireless Sensing Smartwatch Application for the Self-Management of Pediatric Asthma. *Sensors* 17, 8 (2017). <https://doi.org/10.3390/s17081780>
- [2] Scott D. Lowther, Sani Dimitroulopoulou, Kerry Foxall, Clive Shrubsole, Emily Cheek, Britta Gadeberg, and Ovnair Sepai. 2021. Low Level Carbon Dioxide

- Indoors—A Pollution Indicator or a Pollutant? A Health-Based Perspective. *Environments* 8, 11 (2021). <https://doi.org/10.3390/environments8110125>
- [3] Nina Sakhnini, Ja Eun Yu, Rachael M. Jones, and Debaleena Chattopadhyay. 2020. Personal Air Pollution Monitoring Technologies: User Practices and Preferences. In *HCI International 2020 - Late Breaking Papers: User Experience Design and Case Studies*, Constantine Stephanidis, Aaron Marcus, Elizabeth Rosenzweig, Pei-Luen Patrick Rau, Abbas Moallem, and Matthias Rauterberg (Eds.). Springer International Publishing, Cham, 481–498.
- [4] Aku Visuri, Zhanna Sarsenbayeva, Niels van Berkel, Jorge Goncalves, Reza Rawasizadeh, Vassilis Kostakos, and Denzil Ferreira. 2017. Quantifying Sources and Types of Smartwatch Usage Sessions. In *Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems* (Denver, Colorado, USA) (*CHI '17*). Association for Computing Machinery, New York, NY, USA, 3569–3581. <https://doi.org/10.1145/3025453.3025817>
- [5] Sailin Zhong, Hamed S. Alavi, and Denis Lalanne. 2020. Hilo-Wear: Exploring Wearable Interaction with Indoor Air Quality Forecast. In *Extended Abstracts of the 2020 CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) (*CHI EA '20*). Association for Computing Machinery, New York, NY, USA, 1–8. <https://doi.org/10.1145/3334480.3382813>